

Stochastic Metal Enrichment in AuroraLF

$z=12.5$ full UVLF and burst comparison

Presenter: Zhu Hourui

AuroraLF

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What changed

Existing UVLF path

$$\text{SFR}(t_i) = \frac{f_b f_*(M_{h,i}) \dot{M}_{h,i}}{10^9},$$
$$f_*(M_h) = \frac{2\epsilon_0}{(M_h/M_c)^{-\beta_*} + (M_h/M_c)^{\gamma_*}},$$
$$L_\nu = L_{\nu,\text{can}} + L_{\nu,\text{TH}}.$$

New diagnostic path

- One-zone gas and metal reservoir on fixed MAH/SFR tracks.
- Records gas metallicity and birth metallicity.
- Birth metallicity gates top-heavy source-time SSP use.
- No feedback into SFR, dust, or HMF weights.

$$L_{\nu,\text{can}}(t_{\text{obs}}) = \int (1 - l_{\text{TH}}) \text{SFR}(t') \ell_{\nu,\text{can}}(t_{\text{obs}} - t') dt',$$
$$L_{\nu,\text{TH}}(t_{\text{obs}}) = \int l_{\text{TH}} \text{SFR}(t') \ell_{\nu,\text{TH}}(t_{\text{obs}} - t') dt'.$$

Mode definitions

$$l_{\text{TH}} = 0 \quad \text{canonical,}$$
$$l_Z = \mathbf{1}[Z_{\text{birth}}(t') \leq 0.05 Z_{\odot}],$$
$$l_{\text{TH}} = \mathbf{1}[z(t') \geq z_{\text{TH}}] l_Z,$$
$$l_{\text{TH}} = \mathbf{1}[z(t') \geq z_{\text{TH}}] l_Z \mathbf{1}\left[\frac{M_h}{\dot{M}_h} \leq \tau_{\text{grow}}\right].$$

Run settings

- $z_{\text{TH}} = 10$
- $\tau_{\text{grow}} = 50 \text{ Myr}$
- $Z_{\text{birth,TH}} \leq 0.05 Z_{\odot}$
- top-heavy SSP metallicity = $0.05 Z_{\odot}$

One-zone metal update

$$M_{\text{gas},i} = f_{\text{gas}} f_b M_{h,i},$$

$$Z_{\text{birth},i}/Z_{\odot} = \frac{M_{Z,i-1}}{M_{\text{gas},i} Z_{\odot}} 10^{\mathcal{N}(0, \sigma_{\text{birth}})},$$

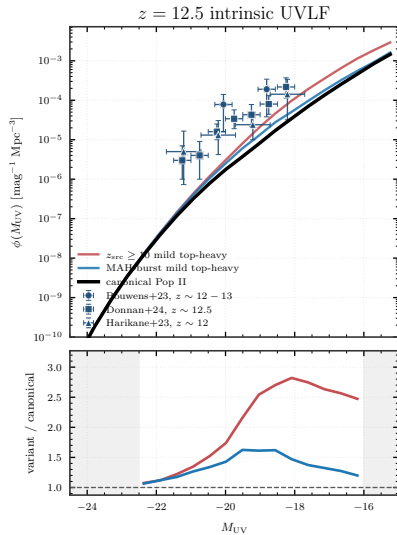
$$\Delta M_{\star,i} = \text{SFR}_i \Delta t_i 10^9,$$

$$\eta_i = \eta_0 \left(\frac{M_{h,i}}{M_{\eta}} \right)^{\alpha_{\eta}} \left(\frac{1+z_i}{z_{\eta}} \right)^{\beta_{\eta}} 10^{\mathcal{N}(0, \sigma_{\eta})},$$

$$y_i = y \left[1 + (m_{\text{TH}} - 1) l_{\text{TH},i} \right] 10^{\mathcal{N}(0, \sigma_y)}.$$

$$M_{Z,i} = \max \left[M_{Z,i-1} + Z_{\text{in}} \Delta M_{\text{in},i} + y_i \Delta M_{\star,i} - M_{Z,i-1} (1 - e^{-[(1-R)+\eta_i] \Delta M_{\star,i} / M_{\text{gas},i}}), 0 \right].$$

$z = 12.5$ UVLF comparison



Burst scatter model

Implemented perturbation

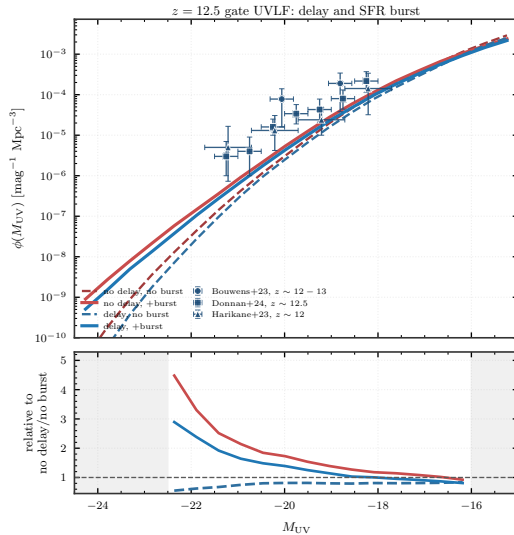
$$\text{SFR}_{\text{burst}}(t_i) = \text{SFR}_{\text{smooth}}(t_i) 10^{\Delta_i},$$
$$\Delta_i \sim \mathcal{N}\left[-\frac{1}{2} \ln(10) \sigma_b^2, \sigma_b\right].$$

- Mean-preserving in linear SFR.
- Piecewise correlated over τ_b .
- Same burst realization across IMF modes.

$z = 12.5$ test

- $\sigma_b = 0.5$ dex
- $\tau_b = 20$ Myr
- seed = 777
- metallicity evolution uses burst SFR
- UV convolution uses burst SFR

$z = 12.5$ gate: delay and burst



$z = 12.5$ run summary

Run

- $N_{\text{mass}} = 3000$, $n_{\text{tracks}} = 1000$
- HMF: Reed07 via hmf
- stochastic metallicity enabled
- $y = 0.02$, $m_{\text{TH}} = 1.28$
- $Z_{\text{birth,TH}} \leq 0.05 Z_{\odot}$

Result

Mode	UVLF ratio	$f_{\text{TH,src}}$	$Z_{\text{gas,final}}$
z-gate	1.87	0.627	0.117
MAH-burst	1.23	0.412	0.115
canonical	1.00	0.000	0.114

UVLF ratios are medians over model bins; metallicities are medians in $Z/Z_{\odot}$.

Burst run summary

Burst / no-burst UVLF

Mode	median	min	max
canonical	1.59	1.07	4.25
z-gate	1.46	0.92	4.48
MAH-burst	1.35	1.00	4.41

Numbers use $-22.5 \leq M_{\text{UV}} \leq -16$.

Burst run IMF ratios

Mode	ratio	$f_{\text{TH,src}}$
z-gate	1.80	0.669
MAH-burst	1.16	0.440
canonical	1.00	0.000

Canonical ϕ median rises from 2.75×10^{-6} to 4.46×10^{-6} .

Delay and burst matrix

Gate UVLF, relative to no-delay/no-burst

Case	median	min	max
no delay, +burst	1.46	0.92	4.48
delay, no burst	0.80	0.55	0.84
delay, +burst	1.20	0.81	2.89

Numbers use $-22.5 \leq M_{UV} \leq -16$.

Within-model effects

Ratio	median
delay +burst / delay no-burst	1.48
delay +burst / no-delay +burst	0.81
delay no-burst / no-delay no-burst	0.80

Delay suppresses the absolute gate UVLF; burst partly compensates but does not fully recover the no-delay burst case.

- **z-gate mode**: low- Z gate leaves 62.7% of source-times top-heavy.
- In the observed magnitude range, z-gate boosts UVLF by 1.07–2.82.
- **MAH-burst mode**: source fraction falls to 41.2%; UVLF boost is only 1.07–1.62.
- Burst scatter brightens the UVLF; delay convolution suppresses it by $\simeq 20\%$ in the observed range.
- Even with low- Z top-heavy SSPs and burst scatter, the model remains below many $z \simeq 12.5$ points.

References



Reed et al. 2007, halo mass function fitting functions.



Eldridge et al. 2017, BPASS stellar population synthesis.



Stanway & Eldridge 2018, BPASS v2.2 binary populations.



AuroraLF local implementation, `codex/topheavy-stochastic-metallicity`.

The branch now tracks stochastic metal enrichment consistently with the same source-time top-heavy IMF gates used by the UVLF pipeline.

The burst branch adds short-timescale SFR scatter and shows that this helps the absolute UVLF more than the top-heavy relative ratio.